

Absolute-Value and Piecewise Functions — Evaluating

Evaluating absolute-value functions, choosing the correct branch of a piecewise rule, and reading a piecewise model in context

Directions:

- Every problem gives you a rule and an input. Your job is to produce the output.
- With absolute value, finish everything *inside* the bars first, then take the absolute value, then do what is left outside.
- With a piecewise rule, **read the condition before you touch the formula**. Ask which interval the input falls in; only then substitute.
- Show the substitution, not just the answer — the substitution is what a grader can follow.

Function Reference — keep this beside you

$f(x) = |x - 3| + 2$ (a V-shape; its graph is drawn below)

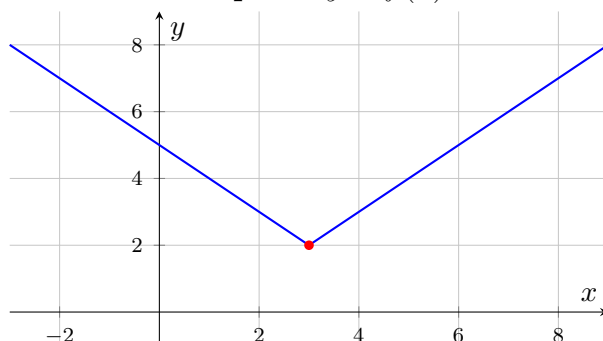
$g(x) = 2|x + 4| - 5$

$$p(x) = \begin{cases} 3x + 1, & x < 2 \\ x^2 - 3, & x \geq 2 \end{cases} \quad q(x) = \begin{cases} -x - 1, & x < -1 \\ x^2, & -1 \leq x < 1 \\ 4 - x, & x \geq 1 \end{cases}$$

$C(t)$ = cost in dollars of parking for t hours at the town garage:

$$C(t) = \begin{cases} 4, & 0 < t \leq 1 \\ 4 + 3(t - 1), & 1 < t \leq 5 \\ 16, & t > 5 \end{cases}$$

Graph of $y = f(x)$



The lowest point of the V sits where the inside of the bars is zero.

1. Use $f(x) = |x - 3| + 2$. Find $f(-1)$.

Write the substitution first, then simplify inside the bars, then finish.

Answer: _____

2. Complete the table of values for the same $f(x) = |x - 3| + 2$. Then mark the three points on the graph in the Function Reference and check that they land on the V.

x	0	3	7
$f(x)$			

$f(0) =$ _____ $f(3) =$ _____ $f(7) =$ _____

3. Use $g(x) = 2|x + 4| - 5$. Find $g(-9)$.

Careful: the 2 multiplies the absolute value, so the bars close before the multiplying starts.

Answer: _____

4. Use the piecewise rule $p(x) = \begin{cases} 3x + 1, & x < 2 \\ x^2 - 3, & x \geq 2 \end{cases}$

(a) Find $p(-2)$. (b) Find $p(2)$.

For each one, name the branch you used and say which condition made you pick it.

$p(-2) = \underline{\hspace{2cm}}$ $p(2) = \underline{\hspace{2cm}}$

5. Use the parking-cost model $C(t)$ from the Function Reference. A driver parks for 4 hours.

How many dollars does the driver pay?

Answer: $\underline{\hspace{2cm}}$

6. Use the three-branch rule $q(x) = \begin{cases} -x - 1, & x < -1 \\ x^2, & -1 \leq x < 1 \\ 4 - x, & x \geq 1 \end{cases}$

Find $q(-3)$, $q(0)$, and $q(1)$. One of these three inputs sits exactly on a boundary — say which one, and which branch its condition sends you to.

$q(-3) = \underline{\hspace{2cm}}$ $q(0) = \underline{\hspace{2cm}}$ $q(1) = \underline{\hspace{2cm}}$

7. Look again at the graph of $f(x) = |x - 3| + 2$ in the Function Reference. The horizontal line $y = 6$ crosses the V twice.

Find *both* inputs x for which $f(x) = 6$. Solve it algebraically, then check your two answers against the picture.

Answer: $\underline{\hspace{2cm}}$

$$C(3.5) = \underline{\hspace{2cm}} \quad t = \underline{\hspace{2cm}}$$

Explain in your own words: why do the two formulas disagree at $x = 2$, which value does the definition of p actually assign to $p(2)$, and what does the graph of p do at that input? Use the words *branch* and *condition* in your answer.

Answer: _____

10. Challenge. The two branch formulas of p are $y = 3x + 1$ and $y = x^2 - 3$. Ignore the conditions for a moment and treat them as two ordinary equations.

Find every point (x, y) where the two formulas produce the same output. Then say whether either of those inputs is a place where p itself would be continuous, and why.

Answer: _____